Does network depth track reasoning depth?

N = 24 d_model 256, ff 1024 fixed, only num_layers varies

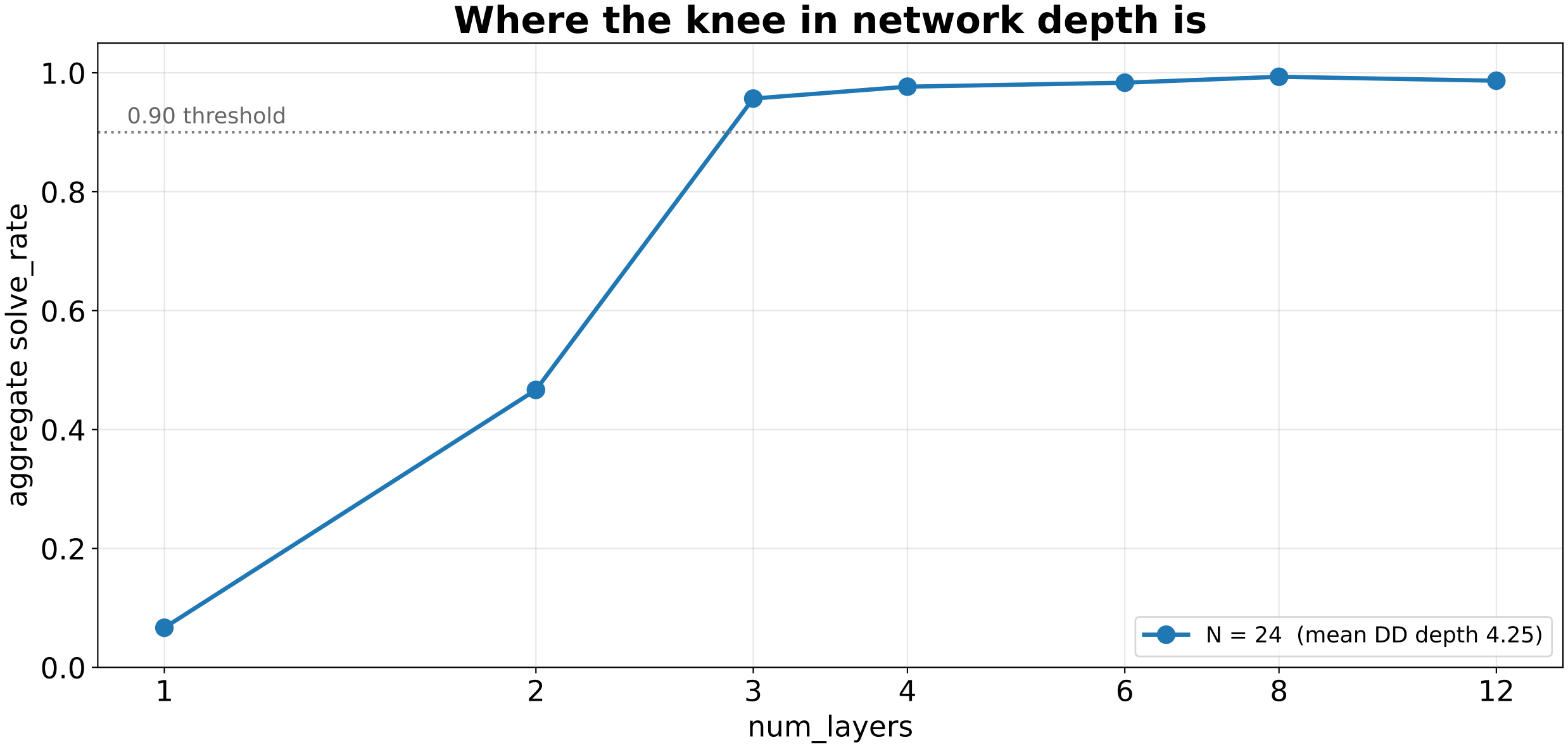
Network depth does NOT track reasoning depth

`N=24, rho=0.75: L*(T) = 3 at every T in 1..3  ·  N=24, rho=0.60: L*(T) = 3 at every T in 2..5`

aggregate solve_rate by num_layers

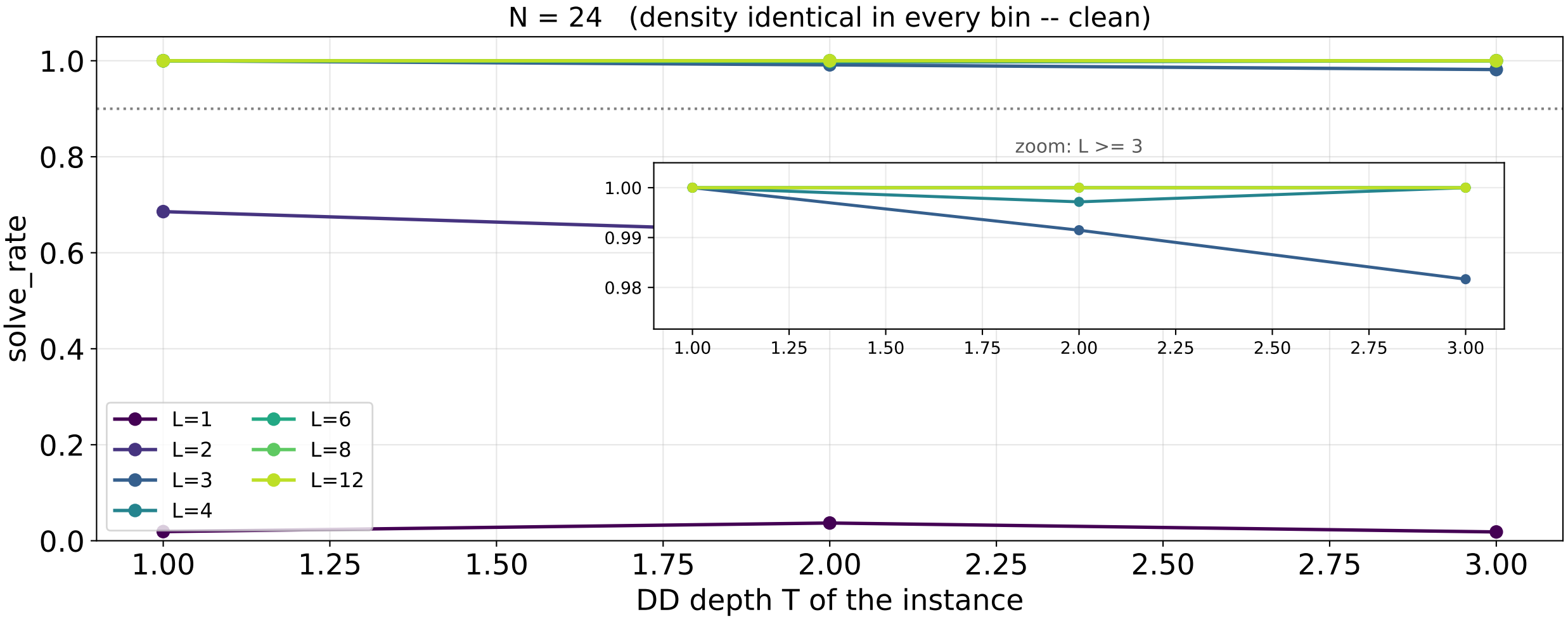
N = 24	L=1	L=2	L=3	L=4	L=6	L=8	L=12
solve	0.067	0.467	0.957	0.977	0.983	0.993	0.987
params (M)	0.86	1.65	2.44	3.23	4.81	6.39	9.55

L*(T) is the smallest num_layers reaching solve_rate >= 0.90 on depth-T instances at fixed hint density. Bins with fewer than 40 instances are excluded from it.

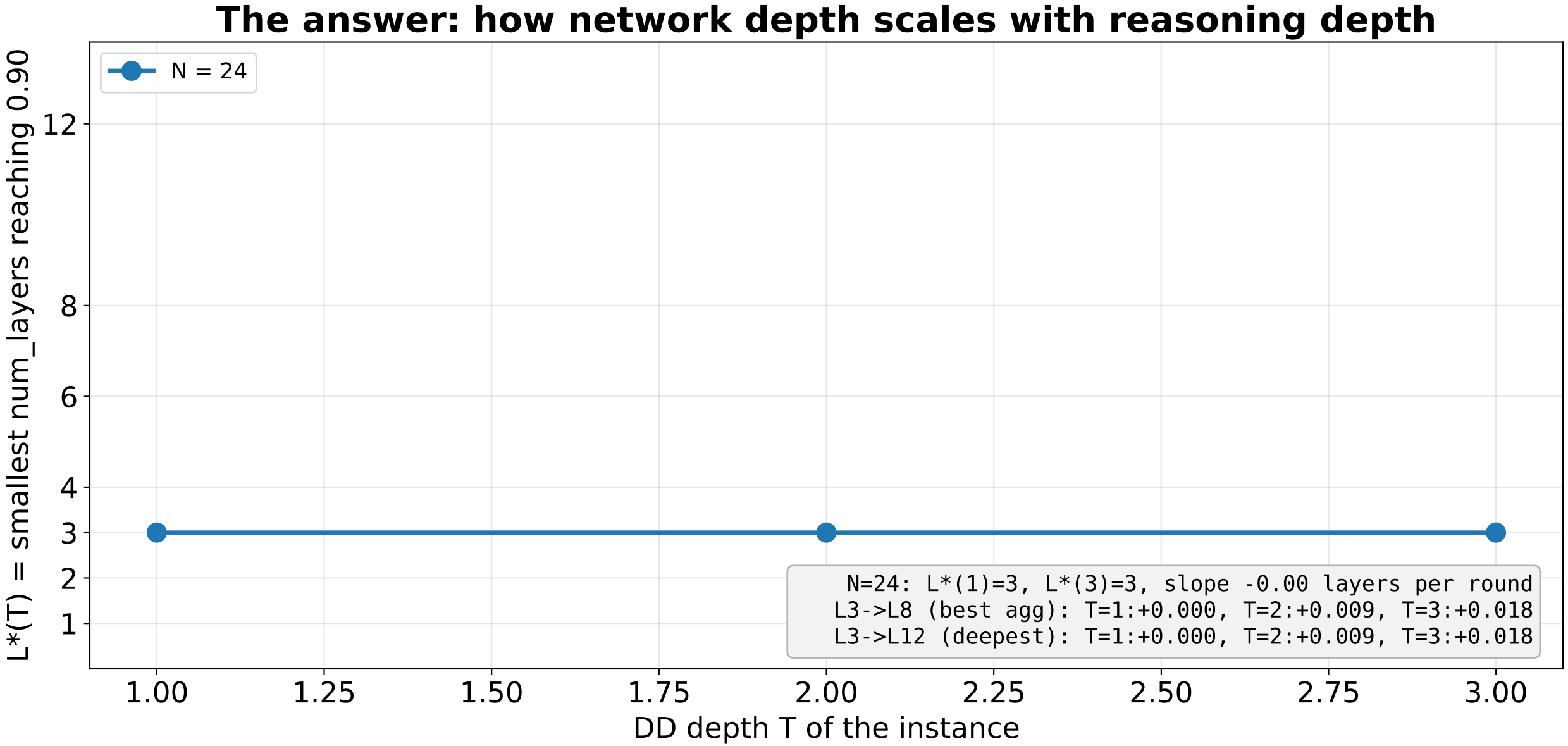


Aggregate over the whole training density range, so it mixes depths and is NOT the answer to the depth question -- it locates the knee that the per-depth pages then test. A knee far below the deepest instance's T is already evidence against one layer per deduction round.

solve_rate by reasoning depth, at fixed density rho = 0.75

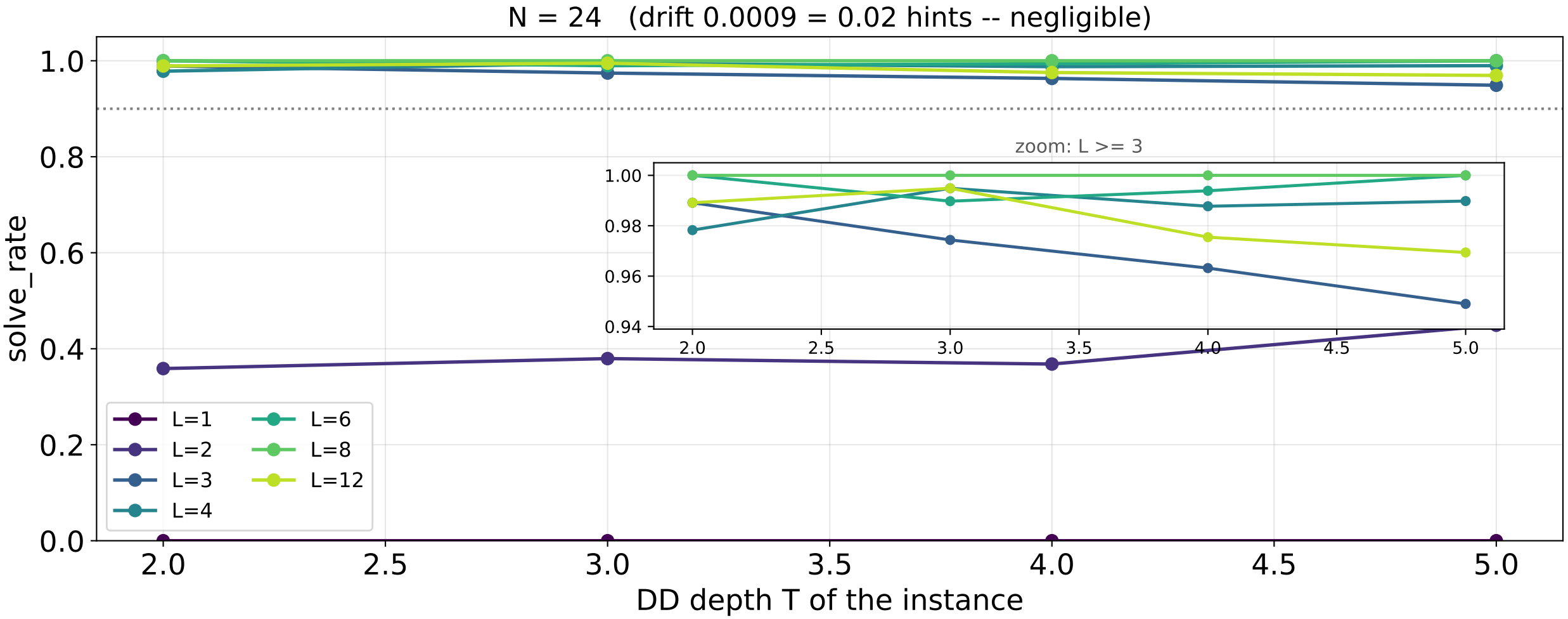


Density is held fixed, so a decline left-to-right is attributable to the length of the deduction chain rather than to how much was given away. The realized rho spread in each title is the check on that: if it is large, deeper-T instances are also sparser and the confound has moved inside the bins. If the curves for different L are PARALLEL, depth buys accuracy uniformly; if they FAN OUT with T, deeper networks reach deeper instances and the serial picture is right.

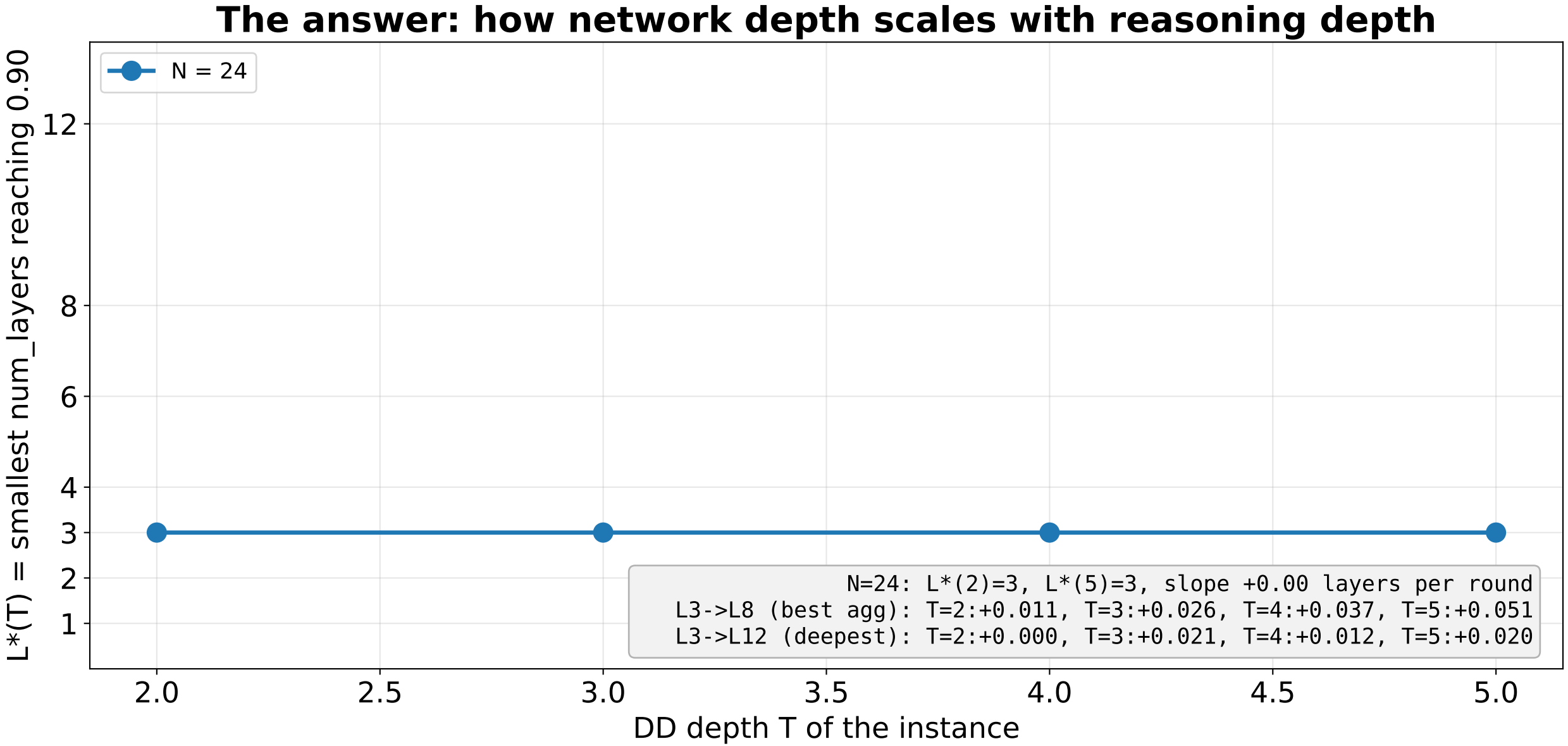


A slope near 0 says the same network depth suffices at every reasoning depth, which is what finding 2 predicts -- the model is not spending one layer per deduction round. A slope near +1 is the serial picture. Crosses mark depths no tested num_layers reached, i.e. L*(T) is off the top of the grid. Measured at rho = 0.75.

solve_rate by reasoning depth, at fixed density rho = 0.60



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Pre-registered predictions, scored

N = 24

(a) T=1 solve >= 0.95 for all L >= 2 `worst is L=2 at 0.686`	**FAILS**
(b) L*(T) rises, L*(max) >= L*(min)+3 [rho=0.75] `L*=[3, 3, 3] over T=[1, 2, 3], rise +0`	**FALSIFIED -> FLAT**
(b) L*(T) rises, L*(max) >= L*(min)+3 [rho=0.60] `L*=[3, 3, 3, 3] over T=[2, 3, 4, 5], rise +0`	**FALSIFIED -> FLAT**
(c) L=1 fails everywhere (< 0.5 at T >= 2) `max over T>=2 bins is 0.037`	**HOLDS**
(d) aggregate saturates by L=8 (|L12-L8| <= 0.01) `L8 0.9933, L12 0.9867, delta 0.0067`	**HOLDS**

(a) failing is the informative one: at T=1 -- a single deduction round -- L=2 still scores 0.686, so the L=3 knee is not about chaining rounds through the stack at all. That is the same conclusion (b) reaches from the other side.